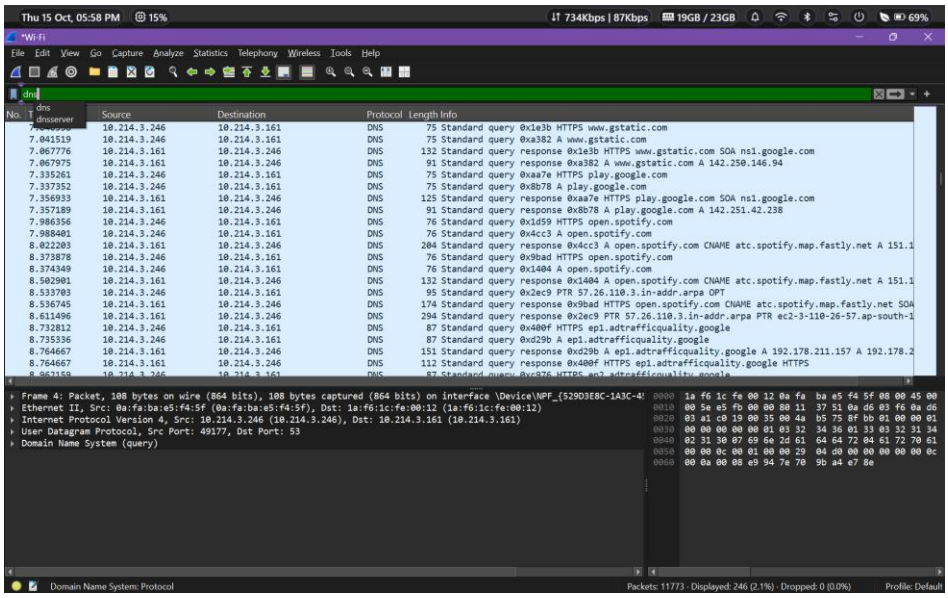
Introduction to Network Protocols

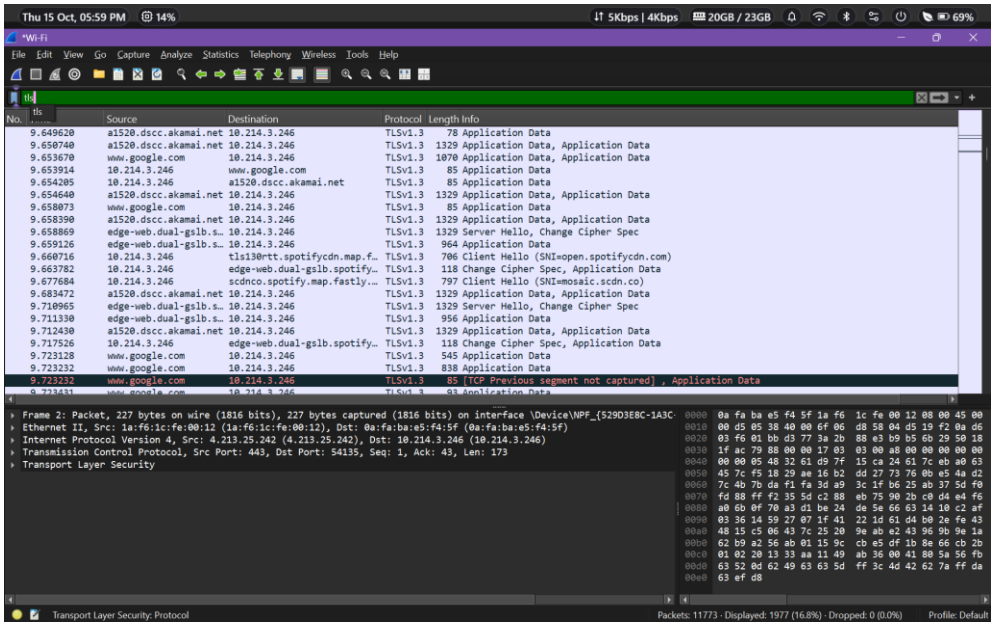
Q1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Solution: Network Traffic Analysis

1. Enter dns in the display filter to view DNS packets.



2. Enter tls to view encrypted HTTPS-related traffic.

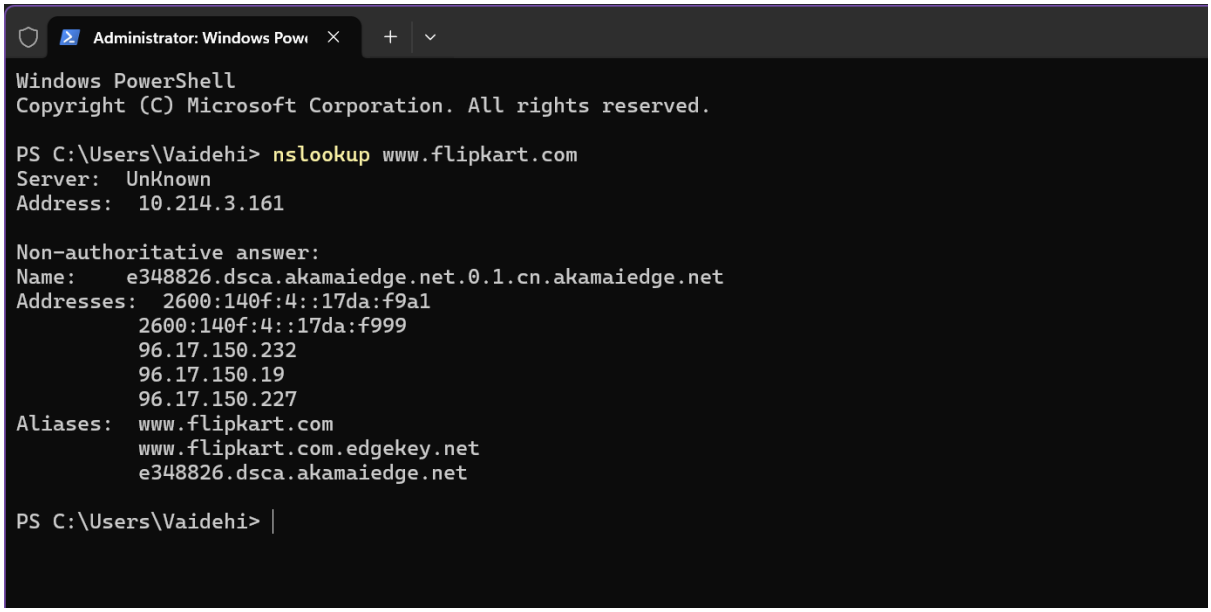


Observations

Packet	Protocol	Information
1	DNS query	Client requests the IP address for www.spotify.com.
2	DNS response	DNS server returns an IP address or a chain of aliases.
3	TCP	Connection is established with a web server.
4	TLS	Encrypted HTTPS communication takes place.

Q2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

Solution: DNS Resolution



Output

Server: unknown

Address: 10.214.3.161

Non-authoritative answer:

Name: e348826.dsca.akamaiedge.net.0.1.cn.akamaiedge.net

Addresses: 2600:140f:4::17da:f9a1

2600:140f:4::17da:f999

The IP address returned by nslookup identifies an address associated with the requested domain name. A website may return multiple IP addresses or aliases, and the result can vary depending on DNS configuration and location.

Q3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

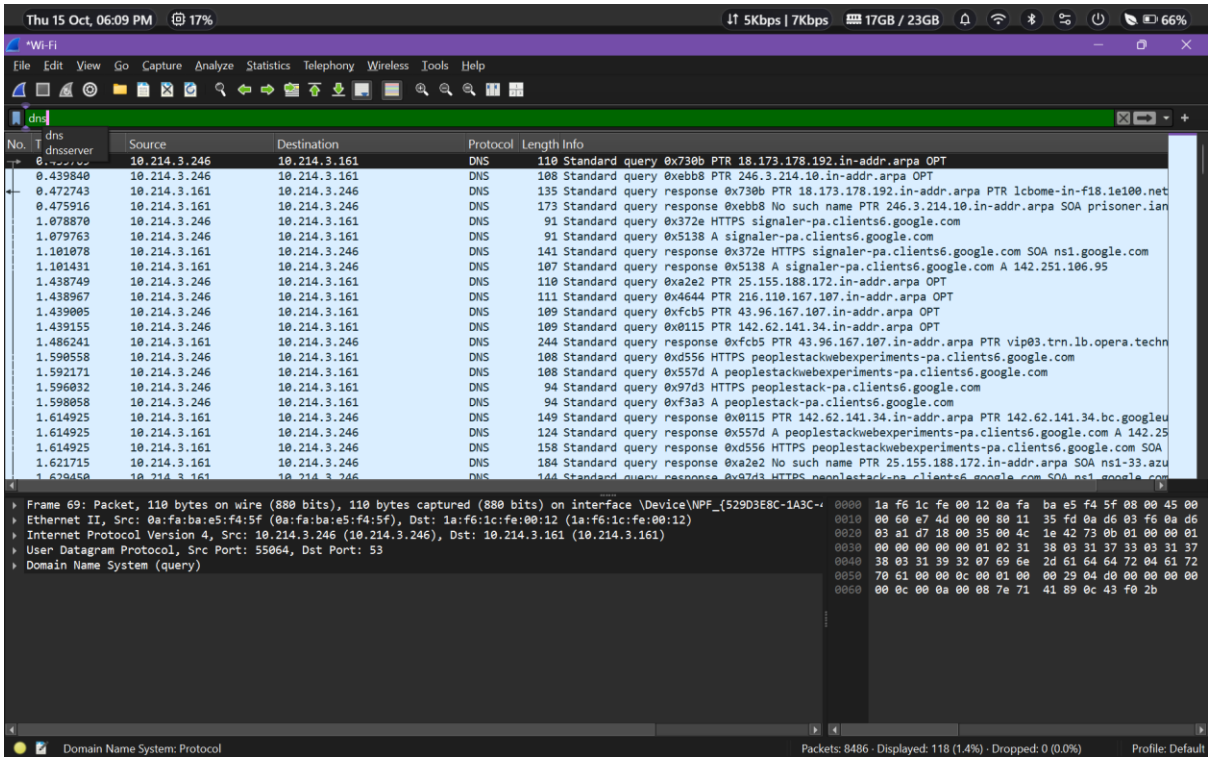
Solution : HTTP is used to transfer web content between a browser and a web server. HTTPS uses HTTP over TLS to encrypt communication.

Feature	**HTTP**	**HTTPS**
Port	80 by default	443 by default
Security	Data is not encrypted by HTTP itself	TLS encrypts data in transit
Authentication and integrity	No TLS certificate protection	TLS helps verify the server and protect data integrity

Real-world example: Instagram uses HTTPS to protect communication between its website or app and its servers, helping safeguard login sessions and other transmitted information.

Q4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe. Hint : Look for packets with 'Standard query' and 'Standard query response' in the Info column.

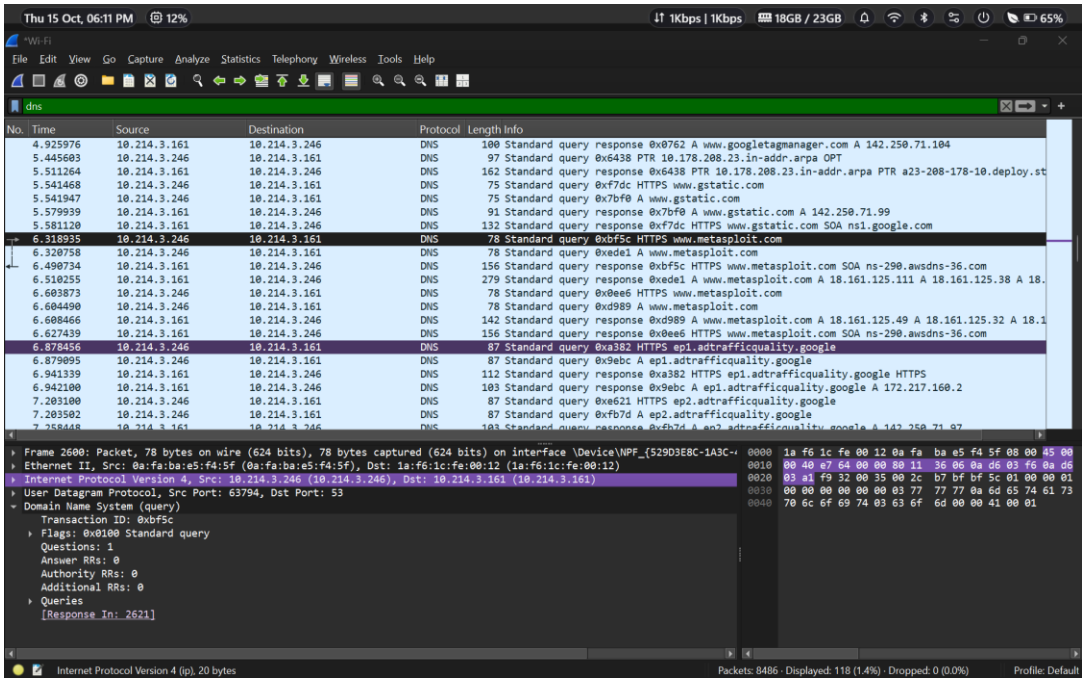
Solution: DNS Traffic Analysis



dns

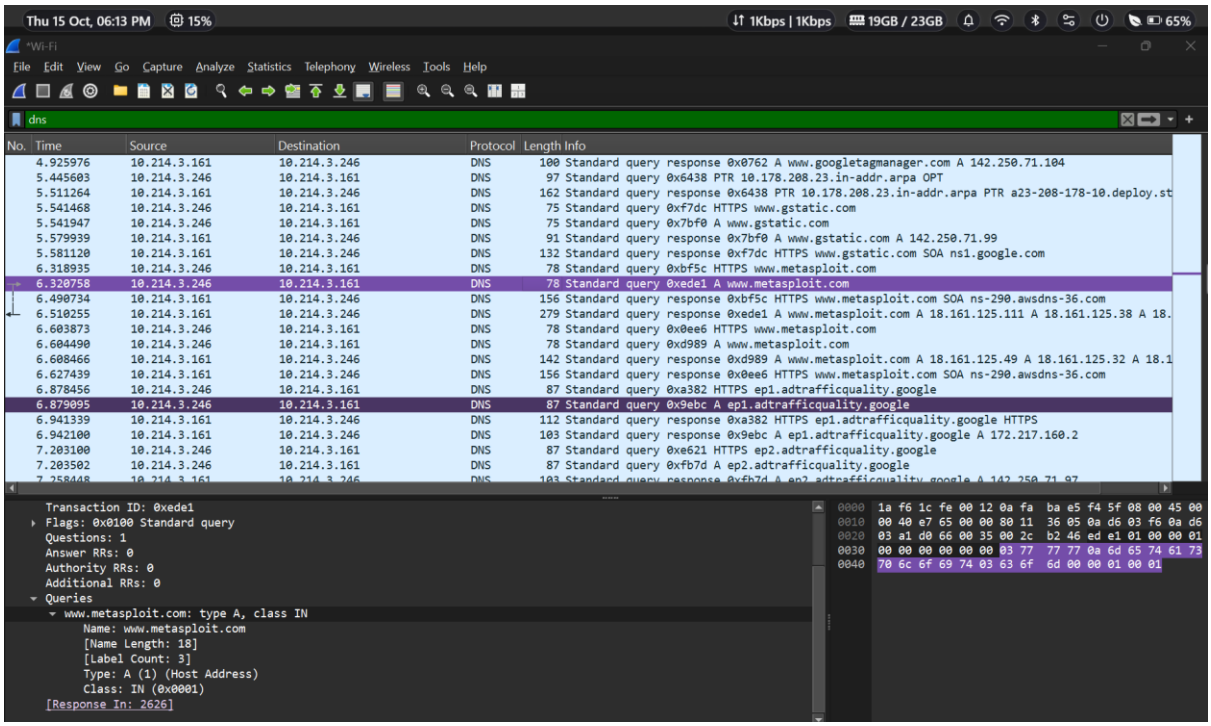
Expected sequence

1. Browser requests a website



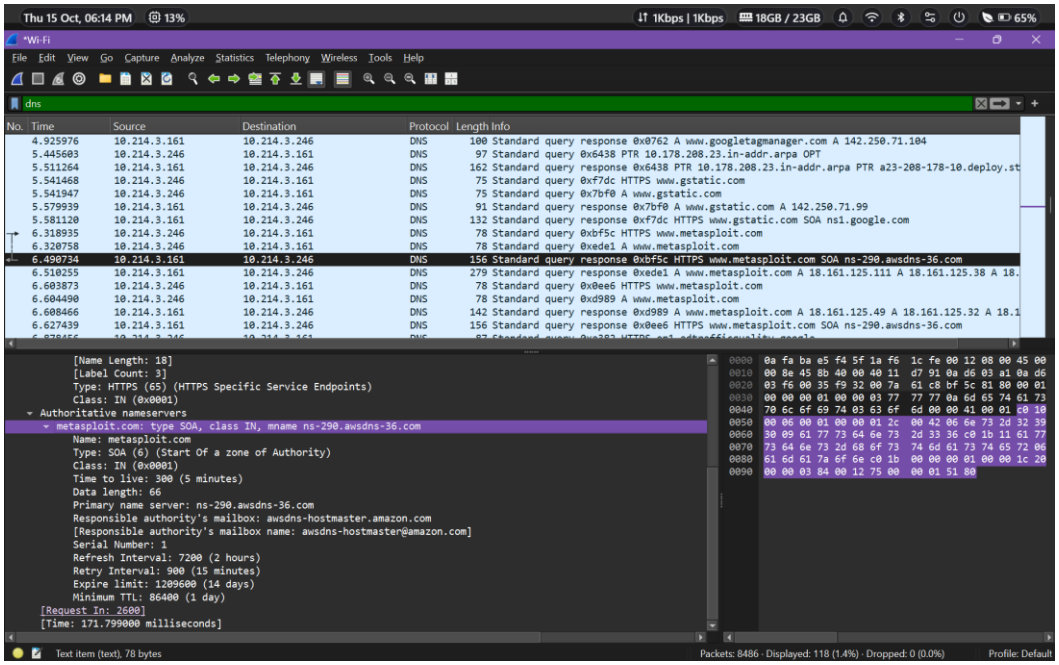
2. DNS Standard query

The client asks for the domain's IP address.



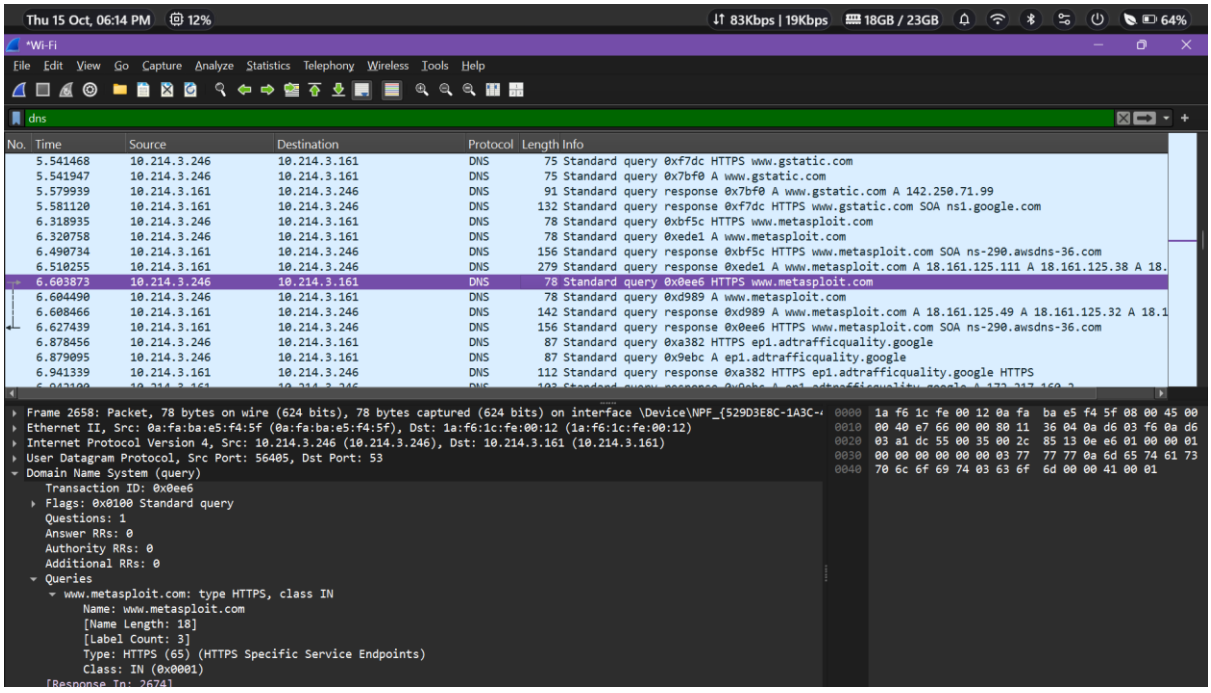
3. DNS Standard query response

The DNS server returns an address or an error.



4. Browser connects to the server

The browser starts web communication using HTTPS in most cases.



DNS queries request domain information, and DNS responses provide the requested record or indicate that it could not be resolved.

Q5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Solution : Real-world use of DHCP

Protocol selected: DHCP (Dynamic Host Configuration Protocol)

When I connect my phone to Wi-Fi at a coffee shop to open Instagram, the router's DHCP service automatically assigns the phone an IP address, subnet mask, default gateway, and DNS server information.

When I connect my phone to a coffee shop's Wi-Fi to use Instagram, DHCP works as follows:

1. Discover: My phone broadcasts a DHCP Discover message to find a DHCP server.
2. Offer: The DHCP server offers an available IP address to my phone.
3. Request: My phone requests the offered IP address.
4. Acknowledgment: The DHCP server confirms the assignment and provides network settings, such as the subnet mask, default gateway, and DNS server.

This allows the phone to communicate on the local network and access the Internet without manually configuring its IP address.